C.A.N.I.S.
Corrosion And Nuclear Inspection System

Store Inspection Report

Report CANIS-preview8_014613 · 2026-11-08

FACILITY	Drum Store 1 (simulated)	RUN ID	`preview8_014613`
STARTED	2026-11-08 01:46:26 UTC	DURATION	00:01:30
INSPECTOR	C.A.N.I.S. (autonomous quadruped)	SOFTWARE	`a1ff9ee`
DETECTOR	YOLO26s, run `yolo26s_20261107T225757Z_d625ef`, cut-off 0.25		
REASONING	Cosmos Reason 2B (context for red drums only)		
STATUS	Simulated store (NVIDIA Isaac Sim). Not for operational use.		

Summary

Drums inspected	**8 / 8**	Drums needing a person	**7**
Red action required	**6**	Areas keep out	**0**
Amber re-inspect	**1**	Areas restricted	**4**
Green no action	**1**	Areas not surveyed	**7**
Robot dose	**0.27 / 1.00 µSv**	Mission time	**00:01:30**

27% of dose budget

Verdict. 6 drums need action; 1 needs re-inspection. 4 aisle sections RESTRICTED for personnel.

1 Access verdict by area

Each aisle is divided into 2 m sections. A section is SAFE only if the robot logged readings there and none of the criteria in Section 6 were met. Sections without readings are NOT SURVEYED, never safe.

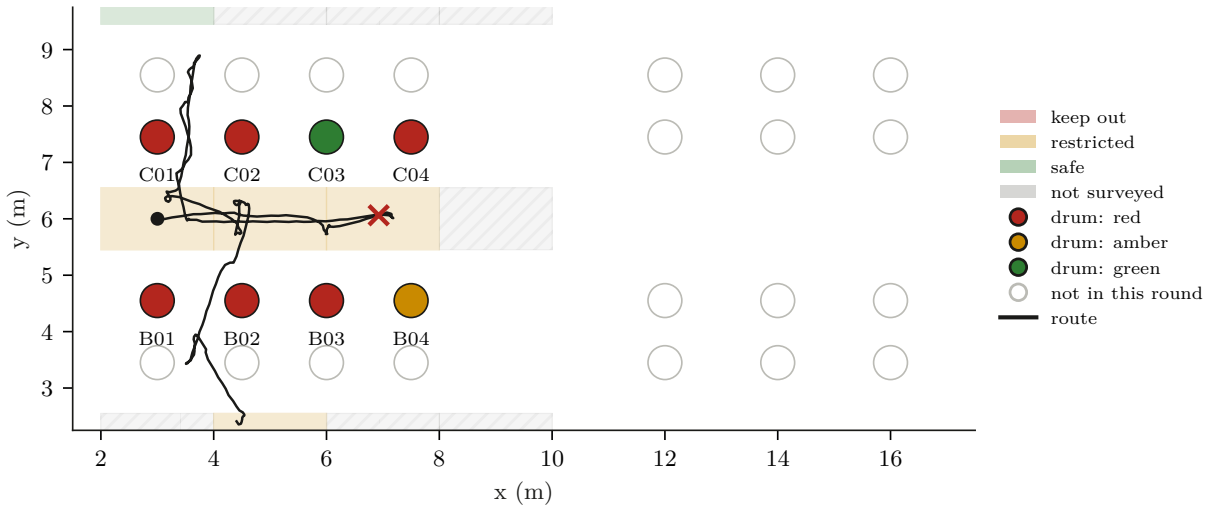


FIGURE 1. Store plan with access zones, drum status and the robot's route.

Aisle	Section	Zone	Basis	Max γ (µSv/h)	H_2 (ppm)
y = 2	2–4 m	NOT SURVEYED	no readings logged	*n/m*	*n/m*
y = 2	4–6 m	RESTRICTED	gamma 7.6 µSv/h at 00:01:21	7.6	*n/m*
y = 2	6–8 m	NOT SURVEYED	no readings logged	*n/m*	*n/m*
y = 2	8–10 m	NOT SURVEYED	no readings logged	*n/m*	*n/m*
y = 6	2–4 m	RESTRICTED	defective drum B01, C01	7.1	*n/m*
y = 6	4–6 m	RESTRICTED	gamma 70.1 µSv/h at 00:00:21	70.1	*n/m*
y = 6	6–8 m	RESTRICTED	gamma 79.3 µSv/h at 00:00:18	79.3	*n/m*
y = 6	8–10 m	NOT SURVEYED	no readings logged	*n/m*	*n/m*
y = 10	2–4 m	SAFE	max gamma 5.7 µSv/h	5.7	*n/m*
y = 10	4–6 m	NOT SURVEYED	no readings logged	*n/m*	*n/m*
y = 10	6–8 m	NOT SURVEYED	no readings logged	*n/m*	*n/m*
y = 10	8–10 m	NOT SURVEYED	no readings logged	*n/m*	*n/m*

2 Drum register

All drums in this round, worst first. Status is set by deterministic rules from the detector and sensor logs; no language model sets any value in this table.

Drum	Cat.	Loc.	Status	Finding	Conf.	Looks	γ µSv/h	Heat Δ°C	Change
B01	LLW	B, x=3	Confirmed defect	bulge+corrosion	0.49	2	7.7	*n/m*	Baseline
B02	LLW	B, x=4.5	Confirmed defect	bulge	0.42	2	25.4	*n/m*	Baseline
B03	LLW	B, x=6	Confirmed defect	bulge+corrosion	0.88	2	77.4	*n/m*	Baseline
C01	LLW	C, x=3	Confirmed defect	bulge+corrosion	0.83	2	8.7	*n/m*	Baseline
C02	LLW	C, x=4.5	Confirmed defect	corrosion	0.82	2	7.0	*n/m*	Baseline
C04	LLW	C, x=7.5	Confirmed defect	corrosion	0.74	2	79.3	*n/m*	Baseline
B04	LLW	B, x=7.5	Suspect	bulge	0.36	1	78.3	*n/m*	Baseline
C03	LLW	C, x=6	OK	none	0.00	1	76.7	*n/m*	Baseline

n/m: not measured in this run. Conf.: highest YOLO confidence for the fault class. Label legibility: unreadable for all drums (camera only).

3 Findings and recommended actions

RED Drum B01 · LLW · row B, x = 3 m

No evidence frame logged for this drum.

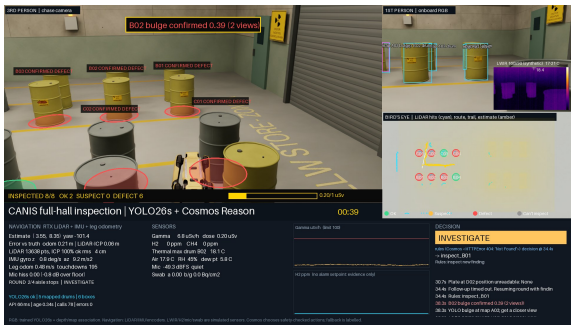
FINDING	bulge+corrosion
CONFIDENCE	0.49 (cut-off 0.25)
VIEWS	corrosion 4, bulge 2 of 40 frames
LOOKS	2
GAMMA	7.7 µSv/h
SWAB	not taken
HEAT	not measured

Basis. YOLO bulge+corrosion in 4 separate views; confirmed bulge+corrosion.

Decision trail.
`00:00:20` B01 corrosion 0.27 (one view) → second look
`00:00:21` B01 corrosion confirmed 0.32 (2 views)
`00:00:34` inspect B01: Rules: inspect new finding
Context (Cosmos Reason). *Not logged in this run.*

Recommended action. Restrict access; arrange an engineering assessment for pressurisation; measure surface temperature (not measured this run).

RED Drum B02 · LLW · row B, x = 4.5 m



Evidence frame at 00:00:39 (confirmed); YOLO boxes as drawn by the detector.

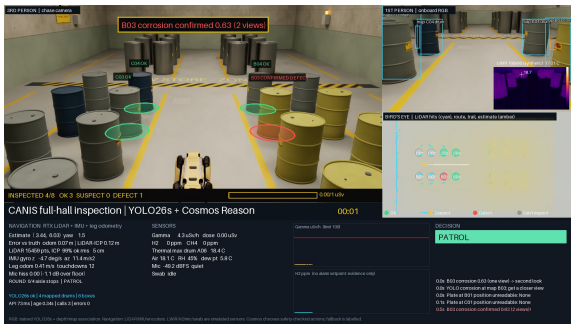
FINDING	bulge
CONFIDENCE	0.42 (cut-off 0.25)
VIEWS	bulge 4 of 30 frames
LOOKS	2
GAMMA	25.4 µSv/h
SWAB	not taken
HEAT	not measured

Basis. YOLO bulge in 4 separate views; confirmed bulge.

Decision trail.
`00:00:21` B02 bulge 0.35 (one view) → second look
`00:00:38` B02 bulge confirmed 0.39 (2 views)
`00:00:53` inspect B02: Rules: inspect new finding
Context (Cosmos Reason). *Not logged in this run.*

Recommended action. Restrict access; arrange an engineering assessment for pressurisation; measure surface temperature (not measured this run).

RED Drum B03 · LLW · row B, x = 6 m



Evidence frame at 00:00:01 (confirmed); YOLO boxes as drawn by the detector.

FINDING	bulge+corrosion
CONFIDENCE	0.88 (cut-off 0.25)
VIEWS	corrosion 26, bulge 2 of 37 frames
LOOKS	2
GAMMA	77.4 µSv/h
SWAB	taken, result not logged
HEAT	not measured

Basis. YOLO bulge+corrosion in 26 separate views; confirmed bulge+corrosion.

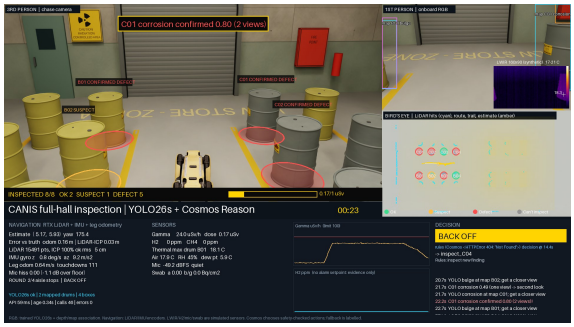
Decision trail.

`00:00:00` B03 corrosion 0.63 (one view) → second look
`00:00:00` B03 corrosion confirmed 0.63 (2 views)
`00:00:05` inspect B03: Rules: inspect new finding
`00:00:07` swab B03: Rules: at standoff, sample for contamination

Context (Cosmos Reason). *Not logged in this run.*

Recommended action. Restrict access; arrange an engineering assessment for pressurisation; measure surface temperature (not measured this run).

RED Drum C01 · LLW · row C, x = 3 m



Evidence frame at 00:00:23 (confirmed); YOLO boxes as drawn by the detector.

FINDING	bulge+corrosion
CONFIDENCE	0.83 (cut-off 0.25)
VIEWS	corrosion 6, bulge 1 of 20 frames
LOOKS	2
GAMMA	8.7 µSv/h
SWAB	taken, result not logged
HEAT	not measured

Basis. YOLO bulge+corrosion in 6 separate views; confirmed bulge+corrosion.

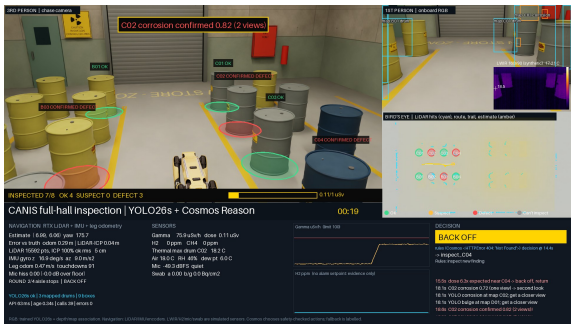
Decision trail.

`00:00:22` C01 corrosion 0.49 (one view) → second look
`00:00:22` C01 corrosion confirmed 0.80 (2 views)
`00:00:43` inspect C01: Rules: inspect new finding
`00:00:46` swab C01: Rules: at standoff, sample for contamination

Context (Cosmos Reason). *Not logged in this run.*

Recommended action. Restrict access; arrange an engineering assessment for pressurisation; measure surface temperature (not measured this run).

RED Drum C02 · LLW · row C, x = 4.5 m



Evidence frame at 00:00:19 (confirmed); YOLO boxes as drawn by the detector.

FINDING	corrosion
CONFIDENCE	0.82 (cut-off 0.25)
VIEWS	corrosion 9 of 12 frames
LOOKS	2
GAMMA	7.0 µSv/h
SWAB	taken, result not logged
HEAT	not measured

Basis. YOLO corrosion in 9 separate views; confirmed corrosion.

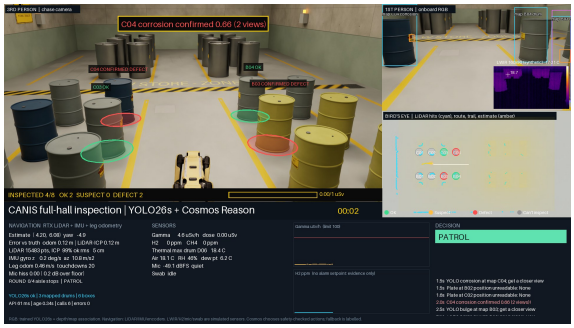
Decision trail.

`00:00:18` C02 corrosion 0.72 (one view) → second look
`00:00:19` C02 corrosion confirmed 0.82 (2 views)
`00:00:56` inspect C02: Rules: inspect new finding
`00:01:01` swab C02: Rules: at standoff, sample for contamination

Context (Cosmos Reason). *Not logged in this run.*

Recommended action. Engineering assessment of container integrity; swab if not taken; consider overpacking.

RED Drum C04 · LLW · row C, x = 7.5 m



Evidence frame at 00:00:03 (confirmed); YOLO boxes as drawn by the detector.

FINDING	corrosion
CONFIDENCE	0.74 (cut-off 0.25)
VIEWS	corrosion 8 of 13 frames
LOOKS	2
GAMMA	79.3 µSv/h
SWAB	not taken
HEAT	not measured

Basis. YOLO corrosion in 8 separate views; confirmed corrosion.

Decision trail.

`00:00:02` C04 corrosion 0.66 (one view) → second look
`00:00:02` C04 corrosion confirmed 0.66 (2 views)
`00:00:14` inspect C04: Rules: inspect new finding
`00:00:16` dose 6.3x expected near C04 → back off, return later

Context (Cosmos Reason). *Not logged in this run.*

Recommended action. Engineering assessment of container integrity; swab if not taken; consider overpacking.

AMBER **Drum B04 · LLW · row B, x = 7.5 m**

No evidence frame logged for this drum.

FINDING	bulge
CONFIDENCE	0.36 (cut-off 0.25)
VIEWS	bulge 1 of 23 frames
LOOKS	1
GAMMA	78.3 µSv/h
SWAB	not taken
HEAT	not measured

Basis. YOLO bulge in only 1 of 23 views; second view did not repeat it; suspect.

Decision trail.
`00:01:18` B04 bulge 0.36 (one view) → second look
Context (Cosmos Reason). *Not logged in this run.*

Recommended action. Re-inspect within 7 days: closer look and swab.

4 Not inspected

- Aisle y = 2, 2–4 m: no readings logged. Treat as unknown until surveyed.
- Aisle y = 2, 6–8 m: no readings logged. Treat as unknown until surveyed.
- Aisle y = 2, 8–10 m: no readings logged. Treat as unknown until surveyed.
- Aisle y = 6, 8–10 m: no readings logged. Treat as unknown until surveyed.
- Aisle y = 10, 4–6 m: no readings logged. Treat as unknown until surveyed.
- Aisle y = 10, 6–8 m: no readings logged. Treat as unknown until surveyed.
- Aisle y = 10, 8–10 m: no readings logged. Treat as unknown until surveyed.

Dose event. At 00:00:15 the gamma rate rose to 40.8 µSv/h (baseline 6.5). The robot backed off at 00:00:16, waited until 00:00:23 and returned at 00:00:26.

5 Self-check against simulation ground truth

The simulation records which drums were really faulty. This file is hidden from the robot and used only here.

Faulty drums in the round	3	
Found (Confirmed or Suspect)	3	
Missed	0	–
False alarms	4	B01, B02, B04, C01
Could not inspect	0	–

6 Method, criteria and limits

YOLO26s	Draws boxes (*drum*, *corrosion*, *bulge*) with a confidence. Cut-off 0.25. Decides nothing.
Rules	Deterministic code turns boxes and sensor readings into drum colours and area zones.
Cosmos Reason 2B	Explains the context of red drums only. Cannot change a status, zone or number; output is checked against the logged facts.
Report	Assembled from logged data only. Missing values are printed as *n/m*.

Criterion	Value	Source
Restricted (gamma)	>7.5 µSv/h	HSE L121 (IRR17 ACOP), reg. 17 paras 297–298; used as an instantaneous robot rule, not a controlled-area assessment.
Keep out (gamma)	>100 µSv/h	HSE L121, guidance on instantaneous dose rates.
Swab	>0.4 / 4 Bq/cm^2	IAEA SSR-6 Rev. 2 (2025) para 508 (α / $\beta\gamma$ non-fixed contamination).
Heat	≥5 °C above air	Team demonstration rule; not an operational limit.
H_2	not set	No cited alarm level. Readings are reported but do not change any zone.

Limits. Simulated store; detector trained on synthetic images; dose from a simplified inverse-square model in simulated units; rust extent estimated, not measured; surface contamination requires a swab; label legibility from the camera only. Structure follows the themes of NDA WPS/640, *Guidance on Monitoring of Waste Packages During Storage*.

A Sensor traces

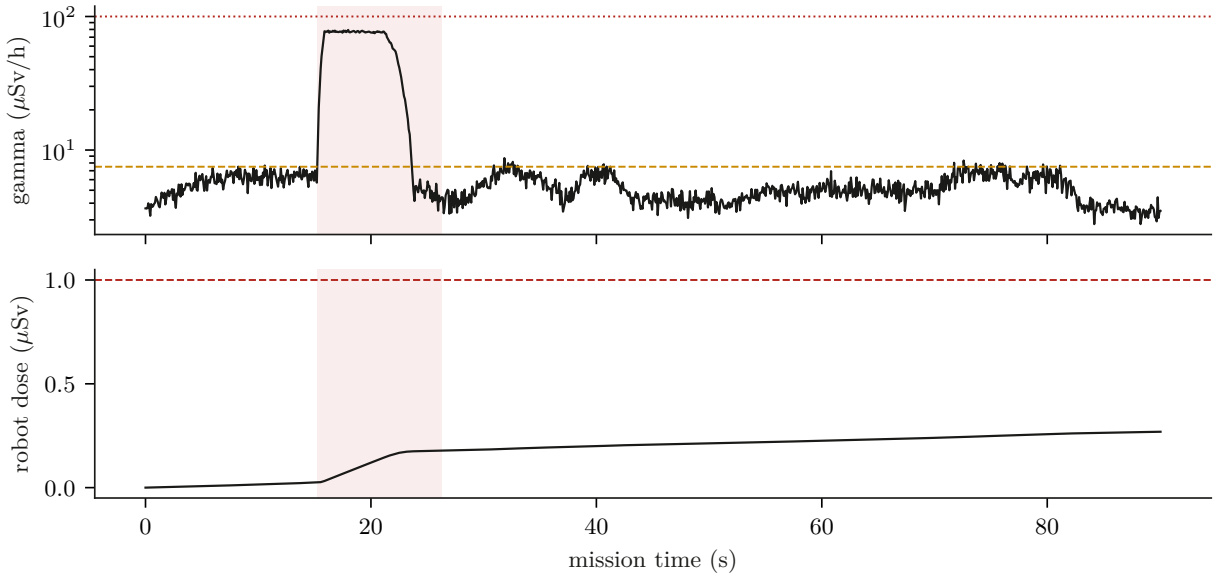


FIGURE 2. Gamma dose rate (dashed: 7.5 µSv/h; dotted: 100 µSv/h), cumulative robot dose (dashed: budget). Shaded: dose event. Air temperature and H_2 were not logged in this run.

B Decision log

Time	Drum	Decided by	Decision
00:00:00	B03	CANIS rule	B03 corrosion 0.63 (one view) → second look
00:00:00	B03	CANIS rule	B03 corrosion confirmed 0.63 (2 views)
00:00:02	C04	CANIS rule	C04 corrosion 0.66 (one view) → second look
00:00:02	C04	CANIS rule	C04 corrosion confirmed 0.66 (2 views)
00:00:05	B03	Rules (Cosmos unavailable)	inspect B03: Rules: inspect new finding
00:00:06	A04	CANIS rule	A04 corrosion 0.50 (one view) → second look
00:00:07	B03	Rules (Cosmos unavailable)	swab B03: Rules: at standoff, sample for contamination

Time	Drum	Decided by	Decision
00:00:14	C04	Rules (Cosmos unavailable)	inspect C04: Rules: inspect new finding
00:00:16	C04	CANIS rule	dose 6.3x expected near C04 → back off, return later
00:00:18	C02	CANIS rule	C02 corrosion 0.72 (one view) → second look
00:00:19	C02	CANIS rule	C02 corrosion confirmed 0.82 (2 views)
00:00:20	B01	CANIS rule	B01 corrosion 0.27 (one view) → second look
00:00:20	D01	CANIS rule	D01 corrosion 0.50 (one view) → second look
00:00:21	B01	CANIS rule	B01 corrosion confirmed 0.32 (2 views)
00:00:21	B02	CANIS rule	B02 bulge 0.35 (one view) → second look
00:00:22	C01	CANIS rule	C01 corrosion 0.49 (one view) → second look
00:00:22	C01	CANIS rule	C01 corrosion confirmed 0.80 (2 views)
00:00:26	D01	Rules (Cosmos unavailable)	better view D01: Rules: low-confidence finding, get a better view
00:00:26	-	CANIS rule	dose back to 4.4 uSv/h → resume round, revisit deferred drums
00:00:34	B01	Rules (Cosmos unavailable)	inspect B01: Rules: inspect new finding
00:00:38	B02	CANIS rule	B02 bulge confirmed 0.39 (2 views)
00:00:43	C01	Rules (Cosmos unavailable)	inspect C01: Rules: inspect new finding
00:00:46	C01	Rules (Cosmos unavailable)	swab C01: Rules: at standoff, sample for contamination
00:00:50	patrol	Rules (Cosmos unavailable)	continue patrol: Rules: sample complete
00:00:53	B02	Rules (Cosmos unavailable)	inspect B02: Rules: inspect new finding
00:00:53	A01	CANIS rule	A01 bulge 0.51 (one view) → second look
00:00:54	A03	CANIS rule	A03 bulge 0.63 (one view) → second look
00:00:55	A03	CANIS rule	A03 bulge confirmed 0.78 (2 views)
00:00:56	C02	Rules (Cosmos unavailable)	inspect C02: Rules: inspect new finding
00:01:01	C02	Rules (Cosmos unavailable)	swab C02: Rules: at standoff, sample for contamination
00:01:03	D01	CANIS rule	D01 bulge+corrosion confirmed 0.50 (2 views)
00:01:05	patrol	Rules (Cosmos unavailable)	continue patrol: Rules: sample complete
00:01:07	A01	Rules (Cosmos unavailable)	inspect A01: Rules: inspect new finding
00:01:18	B04	CANIS rule	B04 bulge 0.36 (one view) → second look
00:01:18	A02	Rules (Cosmos unavailable)	inspect A02: Rules: inspect new finding
00:01:18	A02	CANIS rule	A02 corrosion 0.65 (one view) → second look
00:01:24	A02	CANIS rule	A02 corrosion confirmed 0.65 (2 views)
00:01:25	A02	Rules (Cosmos unavailable)	swab A02: Rules: at standoff, sample for contamination